REINS | one shared SHA-256 compressor

PROPOSAL ONLY

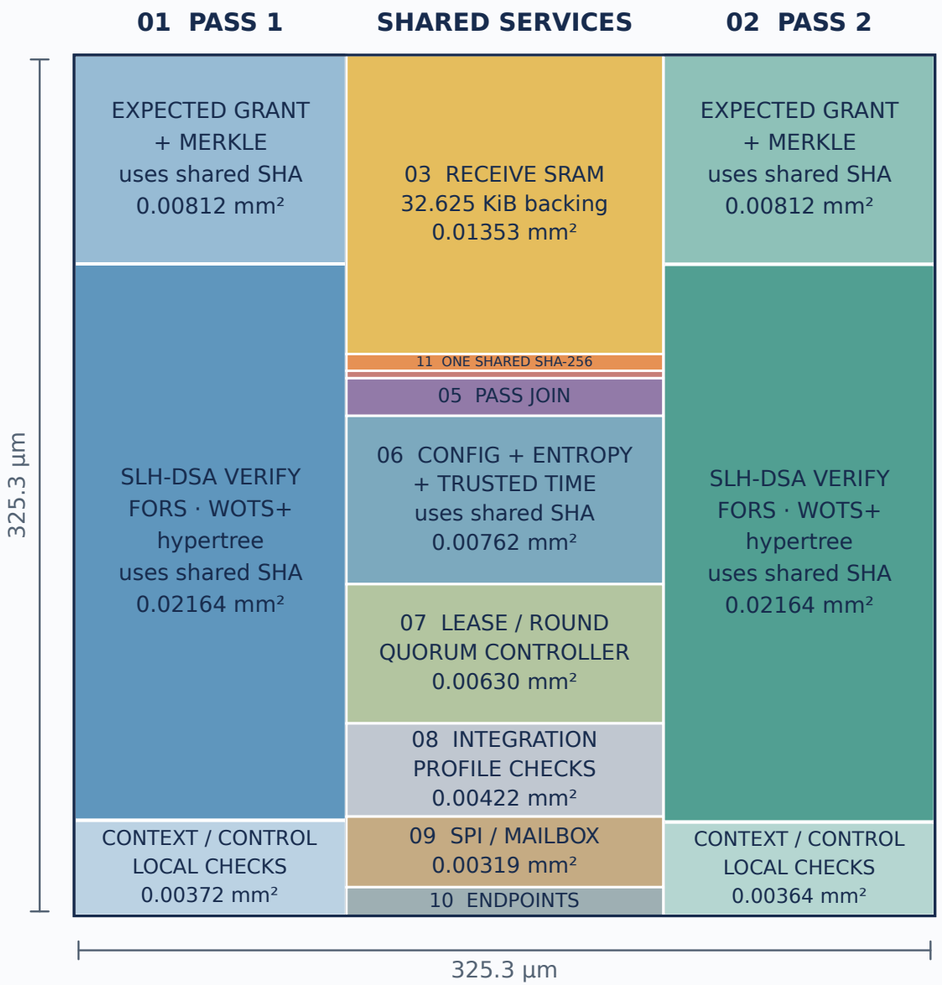
N3 architectural study · area before new sharing logic · not synthesized or placed

2,564 kGE
Nangate45 reference logic

32.625 KiB SRAM
9 KiB payload; backing bits shown

1 SHA-256 engine
Shared across every hash client

0.106 mm²
≈ 325 × 325 µm core



Estimated region areas

Logic regions include placement/routing allowance; SRAM is separate.

REGION	kGE	mm²
01 Verification pass 1	929.89	0.03348
02 Verification pass 2	927.79	0.03340
03 Receive SRAM macro	—	0.01353
11 One shared SHA-256 compressor	21.02	0.00076
04 RX checks / ownership	9.35	0.00034
05 Protected pass join	47.28	0.00170
06 Config / entropy / time	211.53	0.00762
07 Lease / round / quorum	174.98	0.00630
08 Integration / profile	117.09	0.00422
09 SPI / mailbox	88.72	0.00319
10 Endpoint guards / demo	36.11	0.00130
TOTAL DIGITAL CORE	**2,563.77**	**0.10583**

9 → 1 compressors: 5.4% less modeled core area

Both passes and both entropy checks now share a physical fault domain.
Sharing overhead, persistent-fault coverage and timing remain unvalidated.

Planning point: 0.018 µm²/GE · 0.0225 µm²/SRAM bit · logic ×1.25 / 62.5% utilization
SRAM macro = bitcell floor ×2.25. One physical SHA core; all existing framing, scratch and checking retained.

NOT SIZED: OTP (272 bytes), physical entropy sources, timing references/sensors, pads/ESD, power/DFT, and the protected application processor.
NOT INCLUDED: new arbitration, ownership/tag checks, context clearing, routing and any extra fault detection. Production RTL is unchanged.

Estimate = measured nine-core baseline − 8 × 21.0167 kGE bare compressors. No claim of equivalent timing or fault protection.

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